

FCFS Appointment Simulation Results

Supervisor Brief

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Executive Summary

The main result is that capacity utilization and patient access separate sharply in the current FCFS stress-test simulation. At baseline, average utilization is 0.839, but only 0.269 of arrivals complete a visit. This is not a contradiction. Utilization is normalized by the number of appointment slots, while served rate is normalized by arrivals. With 100 arrivals per day and 32 slots per day, the served rate is mechanically bounded near $32/100 = 0.32$, even before no-shows, cancellations, and balking.

The baseline loss pattern is dominated by balking and cancellation, not by a lack of slot offers. Of all arrivals, 26.9% are served, 35.6% balk, 32.3% cancel, 5.2% no-show, and 0.0% receive no offer. The zero no-offer rate matters: at the baseline demand level, the horizon does not fully saturate because daily cancellations keep freeing future capacity. Access losses are therefore mainly behavioral/post-booking losses under long offered delays, not immediate “no slot available” failures.

A controlled no-balking diagnostic shows that eliminating balking does not increase completed utilization. Across 30 seeds, utilization changes from 0.8418 to 0.8404, while booked rate rises by 5.85 percentage points. The additional accepted bookings are offset by no-offer accumulation ($0.0000 \rightarrow 0.2966$) and higher cancellation exposure ($0.3248 \rightarrow 0.3836$). Under the current formulation, balking mainly changes where unmet demand is recorded; it does not solve the completed-visit bottleneck.

The sensitivity analysis gives a clear hierarchy of mechanisms. Total arrival rate is the dominant driver of served rate and offered wait. No-show behavior is the most direct driver of completed utilization because no-show slots are not rebooked. Balking affects who receives access through the shared FCFS pool, but has little direct effect on aggregate utilization near the baseline. Accepted wait is a selective metric because high-delay patients are more likely to balk; offered wait is a better congestion measure, but it must be reported with no-offer share because patients with no offer are absent from all wait metrics.

Bottom line. For this stress-test baseline, the most defensible access summary is not utilization alone. It should report served rate, offered wait, no-offer share, and the arrival outcome decomposition together. The current simulation results describe the mechanics of this model under severe capacity pressure; they should not be interpreted as calibrated clinic estimates.

1 Takeaway 1: Utilization Is Not Access

The baseline demand-to-capacity ratio is the key fact: the model sends 100 arrivals per day into 32 available daily slots. This is a deliberate stress test. It is useful for understanding metric behavior, but it is not a calibrated clinic setting.

$$\text{average utilization} = \frac{\text{completed visits}}{\text{available slots}}, \quad \text{served rate} = \frac{\text{completed visits}}{\text{arrivals}}.$$

Because the denominators differ, utilization can remain high while access is poor. The service system can fill most slots and still serve only a small share of patients when demand is several times larger than supply. At baseline, the maximum possible served rate is near 0.32; the observed value is 0.269 after no-shows and cancellations. This is why utilization alone would be a misleading summary of patient access in this scenario.

Metric	Baseline value	Interpretation
Average utilization	0.839	Completed visits per available slot
Overall served rate	0.269	Completed visits per arrival
Mean offered wait	9.30 days	Delay among patients receiving an offer
Mean accepted wait	8.35 days	Delay among patients accepting the offer
Overall balking rate	0.356	Offered patients who reject long delays
Class served-rate gap	0.001	Near zero under symmetric class parameters

Table 1: Baseline FCFS stress-test metrics from the main note.

2 Takeaway 2: The Main Losses Are Balking and Cancellation

Every arrival is assigned to one outcome category. The baseline decomposition shows that most losses occur after a slot is offered or booked. No-offer is zero, so the baseline failure mode is not horizon saturation.

Outcome	Share of arrivals	Main interpretation
Served	0.269	Completed visit
Balked	0.356	Received an offer, rejected the delay
Canceled	0.323	Booked, then canceled before service
No-show	0.052	Booked, did not attend
No-offer	0.000	Horizon full before any offer
Unresolved	< 0.001	Not resolved before simulation end

Table 2: Baseline arrival outcome decomposition.

The cancellation share is high, but it is mechanically consistent with the model’s daily cancellation rule and long booking delays. With a mean accepted wait of about 8.35 days and a 10% daily cancellation probability, a rough survival calculation gives $0.9^{8.35} \approx 0.416$, or an implied cumulative cancellation risk near 58% among long-wait bookings. The observed cancellation share among booked patients is therefore directionally consistent with the implementation.

The important interpretation is not that cancellation is empirically realistic. It is that long booking delays compound daily cancellation exposure, and the simulation is behaving consistently with that assumption.

3 Takeaway 3: Removing Balking Does Not Recover Capacity

The cleanest diagnostic in the note is the 30-seed comparison between the baseline and a no-balking counterfactual. If balking were the main capacity bottleneck, suppressing it should increase completed utilization. It does not.

Metric	Baseline	No-balking	Change
Average utilization	0.8418	0.8404	−0.0014
Overall served rate	0.2692	0.2682	−0.0011
Booked rate	0.6450	0.7034	+0.0585
Balked share	0.3550	0.0000	−0.3550
Canceled share	0.3248	0.3836	+0.0588
No-offer share	0.0000	0.2966	+0.2966
Mean accepted wait	8.33 d	9.45 d	+1.12 d

Table 3: Controlled no-balking diagnostic over 30 seeds.

The mechanism is a shift in accounting. Without balking, more patients accept appointments, but those bookings extend farther into the horizon, increase cancellation exposure, and create no-offer arrivals once the horizon fills. Completed visits remain essentially unchanged. This is why balking should be treated as an access-allocation and loss-channel mechanism, not as a direct aggregate utilization lever.

4 Takeaway 4: Driver Hierarchy and Reporting Implications

The one-at-a-time sensitivity slices and randomized regression screen are consistent. They identify different mechanisms rather than one universal “best” metric.

Result target	Strongest evidence	Interpretation
Overall served rate	Arrival rate coefficient $\beta = -0.774$	Demand pressure dominates access when capacity is fixed.
Mean offered wait	Arrival rate coefficient $\beta = +0.576$	More demand pushes offers farther into the booking horizon.
Utilization	No-show threshold coefficients +0.342, +0.366; no-show step coefficients −0.289, −0.294	No-show behavior directly changes completed visits because missed slots are not rebooked.
Balking effect	Balking-step utilization effect near zero in OAT; separate regression $\beta = +0.016$, $p = 0.72$	Balking reallocates access and loss channels more than it changes aggregate slot completion.
Class access gap	Behavioral parameters have opposite-sign own-class and cross-class effects	FCFS is neutral in rule form, but shared capacity transmits behavioral differences into class-level access gaps.

Table 4: Highest-signal sensitivity findings from the main note.

Two reporting choices follow directly from these results. First, offered wait should be preferred over accepted wait when diagnosing congestion because offered delay is recorded before the balking decision. Accepted wait excludes patients who rejected high-delay offers, so it can appear to improve exactly when access is worsening. Second, offered wait should always be paired with no-offer share. Once the horizon saturates, no-offer patients receive no delay information and are excluded from wait metrics entirely.

5 Caveats for Interpretation

- The baseline is a stress test, not a calibrated clinical scenario. The 3.1:1 demand-to-capacity ratio is intentionally severe.
- The findings describe the mechanics of the current FCFS formulation. They should not be interpreted as causal estimates for real appointment systems.
- The symmetric baseline has essentially no class gap, as expected. Class-gap estimates near this baseline are noisy with few seeds; stronger equity conclusions require more replicated simulations.

Where to Dig in the Full Note

Brief topic	Full-note reference
Baseline setup and metric definitions	Sections 2.1–2.3; Appendix A
Utilization vs. served-rate gap	Section 2.3; Section 5
Baseline outcome decomposition	Section 3.1; Table 2
Balking suppression diagnostic	Section 3.2; Table 3; Section 7 for the balking deep dive
Offered vs. accepted wait	Sections 4 and 5.1–5.3
Sensitivity driver hierarchy	Sections 6.1–6.5; Appendix D for the regression table
Class-gap interpretation	Sections 6.5 and 8
Robust findings and caveats	Sections 9.1–9.2; Section 10 for discussion questions